Review of Lithium-Ion Battery Fault Features, Diagnosis Methods, and Diagnosis Procedures

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***Abstract*—The increasing adoption of lithium-ion batteries (LIBs) in low-carbon power systems is driven by their advantages, including long life, low self-discharge, and high-energy density. However, LIB failures degrade performance and cause fire hazards. Effective fault diagnosis is thus critical yet challenging. This article reviews LIB fault mechanisms, features, and methods with object of providing an overview of fault diagnosis techniques, emphasizing feature extraction's critical role in detection via thresholds and isolation via multilevel strategies, and estimating detection quality. Several research gaps exist in current fault diagnosis techniques. Most techniques assume controlled conditions unlike complex real-world systems. Resource limitations often confine analytics to individual fault types, overlooking comprehensive approaches. Furthermore, many sensors lack the capability to detect precursor abnormalities, hampering early fault detection. To address these challenges, we advocate for advanced multiphysics and multiscale methods that incorporate sound, force, and thermal coupling to enhance fault diagnosis robustness in practical, complex systems. Multidimensional feature selection can prevent oversimplification in fault diagnosis. Overall, integrating analytics, sensing, and physics could enable comprehensive multidomain fault diagnosis under actual operating conditions.**

***Index Terms*—Battery fault behavior, battery fault detection, battery fault diagnosis, battery fault estimation, battery fault isolation.**

I. Introduction

RENEWABLE energy sources are garnering increasing attention for achieving carbon peaking and neutrality goals. To enable renewable energy absorption and improve peak load and frequency regulation performance, battery energy storage systems are being widely deployed. Among these systems, lithium-ion batteries (LIBs) are commonly used due to their long cycle life, low-self-discharge rate, and high-energy and power density [1]. However, LIB system failures can cause the performance deterioration and even major dangers, such as fires [2]. Therefore, it is imperative to study the LIB fault behaviors and diagnosis methods. Presently, only a few reviews on LIB fault diagnosis methods exist [3], [4], [5]. Abbreviations and symbols used in this paper are listed in Table I and Table II. This article summarizes the latest LIB fault diagnosis methods from recent years and proposes challenges and research directions for battery fault diagnosis.

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II. Lithium-Ion Battery Fault Behaviors

As shown in Fig. 1, LIB system faults can be classified into battery, battery component, sensor, and actuator faults based on the device where fault occurs. Battery component fault refers to the failure of cooling system, connector, and other equipment [6]. Battery sensor fault refers to voltage sensor fault, current sensor fault, and temperature sensor fault in the battery system. Actuator failures involve issues with the BMS, BTMS, EMS, and etc. [7], [8], [9].

Battery faults are the most studied category. In order to provide a comprehensive understanding of a particular fault diagnosis area, this article focuses on the most typical battery faults, including short circuit fault, thermal runaway (TR), overcharge (OC), and over discharge.

A. Overcharge and Overdischarge

Overdischarge (OD) refers to the battery voltage dropping below the cut-off level during discharge when the state of charge (SoC) is already low [10]. Causes for OC and OD include charger failure [11], BMS failure [12], [13], [14], and battery inconsistency [15]. During discharging, lithium ions are released from the negative electrode and return to the positive electrode, thereby gradually decreasing the potential of positive electrode [16]. As the negative electrode completely delithiates, its potential rises quickly [17]. Continuous OD can thus lead to rapid voltage decrease and may be accompanied with irreversible chemical reactions and internal structural damage [18], [19], resulting in capacity degradation [20] and even internal short circuit (ISC) [21].

OC refers to forcing further charging of the battery after reaching its nominal cut-off voltage [22], [23], [24]. During OC, lithium ion deintercalation increases positive electrode potential. The positive electrode material will be damaged and the resistance will increase sharply, resulting in an increase in the ohmic heat production. Excess delithiation causes lithium to react with electrolyte, thickening the SEI film, which may

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TABLE I
LIST OF ABBREVIATIONS

Item	Description	Item	Description
ANN	Artificial neural network	LSTM	Long short-term memory
BMS	Battery management system	MDM	Mean-difference model
BTMS	Battery thermal management system	MSC	Micro short circuit
CNN	Convolutional neural network	OC	Overcharge
CVR	Charging voltage ranking	OCV	Open circuit voltage
ECM	Equivalent circuit model	OD	Overdischarge
EKF	Extended Kalman filter	PCA	Principal component analysis
EMD	Empirical mode decomposition	P2D	Pseudo-two-dimensional
EMS	Energy management system	RE	Relative error
ESC	External short circuit	SOC	State of charge
HPPC	Hybrid pulse power characterization	SOH	State of health
ISC	Internal short circuit	SVM	Support vector machine
LIBs	Lithium-ion batteries	TR	Thermal runaway
LOF	Local outlier factor	VMD	Variational mode decomposition

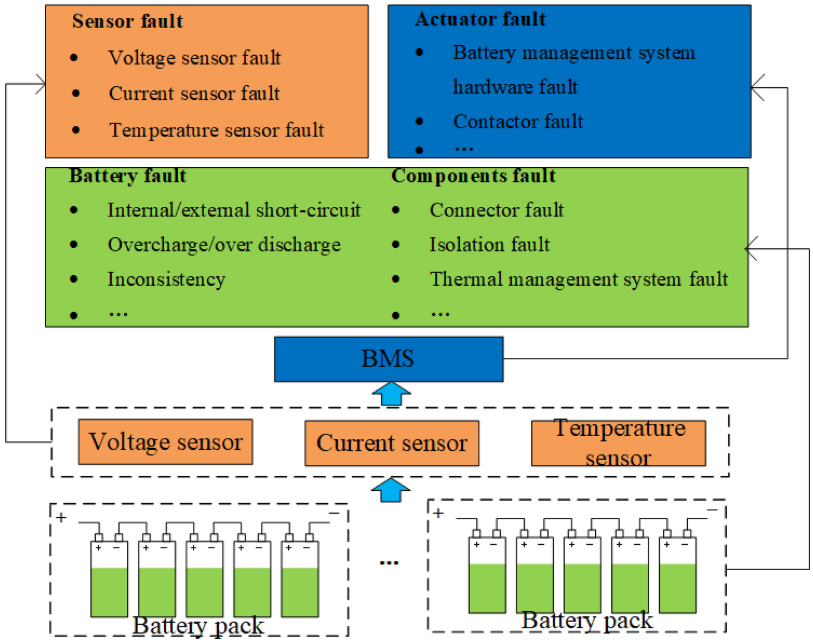


Fig. 1. Classification of battery system fault.

evolve into ISC eventually [25]. OC is a typical and dangerous battery failure mode [26]. ISCs and TRs caused by OC are more severe than OD [27].

B. Short Circuit Faults

Short-circuit faults are the most commonly studied battery failures for diagnosis. When the internal resistance of the battery becomes very low, positive and negative terminal connections of the battery can short circuit for various reasons [28]. A short circuit may cause severe exothermic reactions inside the battery. The generated heat spreads to and affects surrounding cells, triggering further chain exothermic reactions and eventually leading to TR [29].

Based on the location, short circuits are classified as external short circuit (ESC) and ISC [30]. ESC generally refers to the short circuit caused by direct contact between positive and negative electrodes. ESC [31] triggers upon positive and negative battery terminals contact often due to improper operating conditions (such as accidental water or oil leakage into the battery pack or deformation of battery packaging caused by external forces) and system faults (such as loose connecting plate or wire).

During ESC, the battery voltage drops sharply to near-zero and may oscillate [30]. The battery generates high current and heat, leads to ISC in severe cases. ISC is a short circuit occur inside the battery, induced by manufacturing defects, physical abuse and improper operation. Fig. 2 lists several common triggers for ISC [33]. In ISCs, the voltage first drops slowly then rapid drop to zero with heat generation, as depicted in Fig. 3. However, ISCs are difficult to identify initially when voltage and current change gradually [34]. In later stages, significant voltage and temperature changes leave no effective countermeasures to prevent TR.

According to short circuit resistance, battery ISCs can be divided into hard and soft short circuits [35], [36], as shown in Fig. 4. Soft shorts involve high-resistance lithium dendrites. They exhibit voltage drops and local temperature rises from high-short-circuit current densities. At the same time, the lithium metal deposited on the negative electrode reacts with the electrolyte, releasing a lot of heat that melts lithium metal

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TABLE II
LIST OF SYMBOLS

Item	Description	Item	Description
A_k	Coefficient matrix of state space equation	$p(\cdot)$	Probability density of data
α	Quadratic penalty factor	$\boldsymbol{P}_k$	Covariance matrix of EKF
a_p, a_n	Effective solid-liquid interface	p	Data point
$\boldsymbol{B}_k$	Coefficient matrix of state space equation	Q_{irr}	Irreversible power
C_p	Polarization capacitance of ECM	Q_{rev}	Reversible thermal power
$\boldsymbol{C}_k$	Coefficient matrix of state space equation	Q_{ohm}	Heat generating power of ohmic resistance
$c_{s,p}, c_{s,n}$	Concentration of lithium in the solid particle phase	Q_{short}	Short circuit heat
$c_{e,p}, c_{e,n}$	Electrolyte concentration	Q	Total heat
$\boldsymbol{D}_k$	Coefficient matrix of state space equation	R_0	Ohm resistance of ECM
$d_m(\cdot)$	The mth distance of a data point	R_1	Polarization resistance of ECM
$d(\cdot)$	Distance between two points	R_L	Load resistance
$D_{e,p}^{eff}, D_{e,n}^{eff}$	Effective electrolyte diffusivity	R_{SEI}	Resistance of solid electrolyte interface
$D_{s,p}, D_{s,n}$	Solid phase diffusivity	R_{short}	ISC equivalent resistance
$\boldsymbol{E}_k$	Error matrix of EKF at kth step	R	Universal gas constant
η	Heat production efficiency	$R_{s,p}, R_{s,n}$	Solid-phase particles radius
$\varepsilon_p, \varepsilon_n$	Porosity	$r(\cdot)$	Pearson correlation coefficient
F	Faraday constant	$rd_m(\cdot)$	The mth reachable distance between two points
$f(\cdot)$	The sum of mode component after decomposition	$\sigma_{eff,p}, \sigma_{eff,n}$	Effective solid phase conductivity
Φ_s	Solid phase potential	$\kappa_{eff,p}, \kappa_{eff,n}$	Effective liquid phase conductivity
Φ_e	Liquid phase potential	t_+^0	Diffusion constant of lithium ion
γ	Noise tolerance	t	Time
$H(\bullet)$	Shannon entropy	T	Temperature of battery
I_{app}	Applied current	U	Voltage of battery
I_{short}	ISC equivalent current	$\boldsymbol{U}_k$	Input matrix of state space equation at kth step
I_{load}	Load current	U_{OCV}	Open circuit voltage of ECM
j_p, j_n	Pore wall flux	$u_k(\cdot)$	The kth mode component after corresponding decomposition
k_p, k_n	Reaction rate constant	x_i	The ith element of data vector X
$lrd_m(\cdot)$	The mth local reachability density	x_{ave}	Average of vector X
l_p, l_n, l_s	Electrode thickness	$\boldsymbol{X}_k$	State matrix of state space equation
$\boldsymbol{L}_k$	Gain matrix of EKF	$\boldsymbol{X}_k^-$	One step prediction state matrix at kth step
N	Data length	$\boldsymbol{X}_k^+$	State estimation matrix at kth step
$\lambda(\cdot)$	Lagrange multiplicative operator	y_i	The ith element of data vector Y
$N_m(\cdot)$	The mth distance neighborhood	y_{ave}	Average of vector Y
O	Neighbor point of p	$\boldsymbol{Y}_k$	Output matrix of state space equation
ω_k	The kth central frequency after decomposition		

Subscript	Description	Superscript	Description
s, e	Solid phase, electrolyte	eff	Effective
p, n	Positive electrode, negative electrode	n	Step index
i, k, m	Step index	T	Transpose symbol
eff	Effective	-, +	One step prediction, estimation

and the diaphragm. When the temperature continues to rise, soft short circuit evolves into hard short circuit. Differing from soft short circuits, hard short circuits are characterized by low resistance [36], higher short-circuit current density, faster local temperature increases, and higher catastrophic potential like TR [37]. For the external behavior of battery when ISC occurs, ISC voltage may exhibit three behaviors after a significant voltage drop of more than 0.1 V, as shown in Fig. 5 [38]. Behavior I shows that the voltage may drop and recover in a few seconds to a level recognized as soft short circuit; Behavior II shows that the voltage drops to 0 in a second, indicating a hard short circuit; Behavior III shows drops and recovers peaks prior to a final drop, a hard short circuit [38], [39], [40].

Some scholars proposed the concept of micro short circuit (MSC), for shorts with higher resistance (e.g., 100 Ω). Both ISC and ESC could qualify as MSC [41].

C. Thermal Runaway

TR is a phenomenon in which the temperature of a battery cell rises rapidly (millisecond-scale) [42]. TR can result from mechanical abuse, thermal abuse, battery aging, manufacturing defects, OC, and ISC [43]. The sharp temperature rise causes

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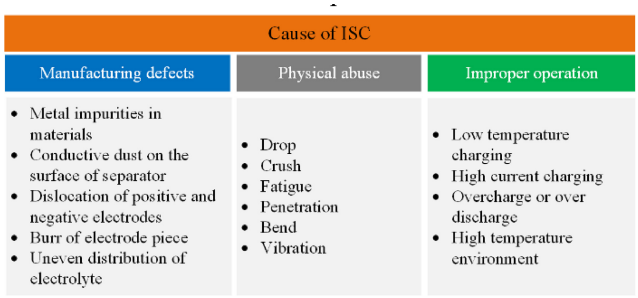


Fig. 2. Triggers of ISC.

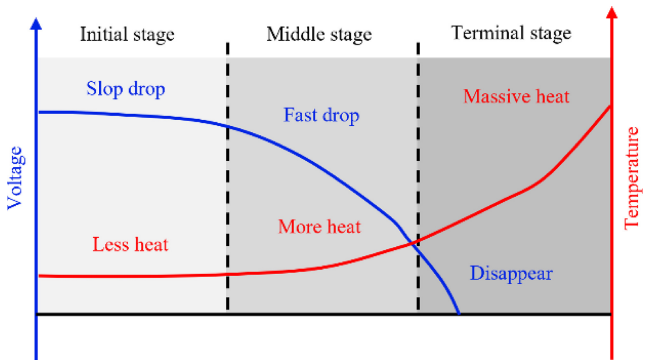


Fig. 3. ISC evolution process.

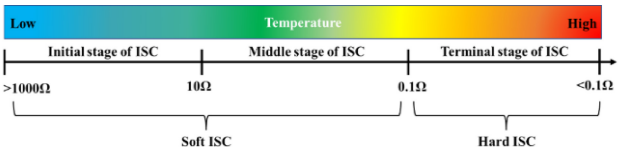


Fig. 4. ISC types according to ISC resistance.

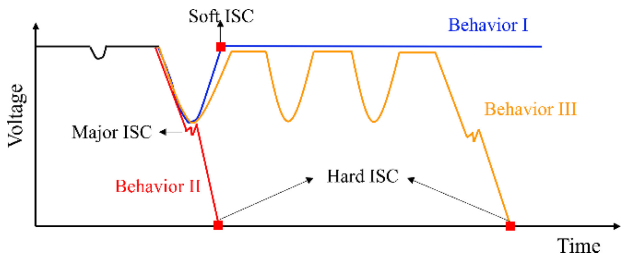


Fig. 5. Different voltage behavior under soft short circuit and hard short circuit [36].

the highly oxidized positive electrode to react with electrolyte, damaging its structure and further heating the cell. The rapid reaction between the positive electrode and the electrolyte triggers a reaction between deposited lithium and the electrolyte, producing significant gas (H_2, CO_2, HF, C_2H_4, etc.), and batteries may undergo sympathetic TR due eventual TR [44]. Additionally, adjacent to the high temperatures of an initial event. This phenomenon is called TR propagation [45]. Temperature behavior under TR is displayed in Fig. 6.

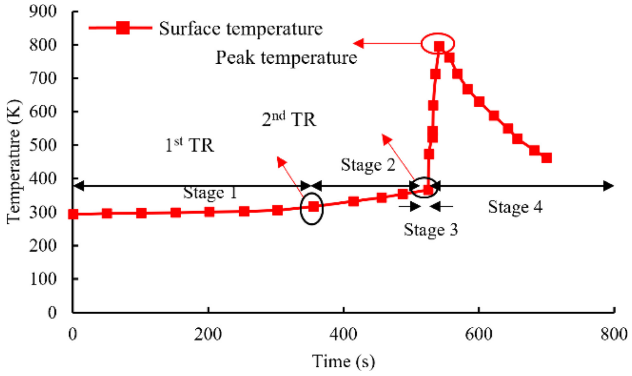


Fig. 6. Temperature behavior under TR [42].

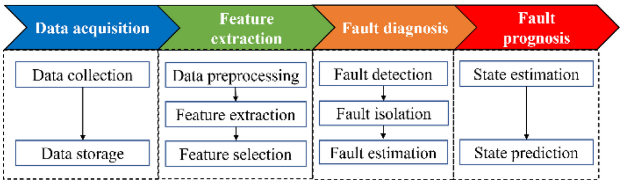


Fig. 7. General LIB fault diagnosis system [4].

III. Lithium-Ion Battery Fault Diagnosis System

The LIB fault diagnosis system generally comprises four steps, as Fig. 7 shows: 1) data acquisition, 2) feature extraction; 3) fault diagnosis; 4) fault prognosis. First, data, such as voltage, current, and temperature of the battery system, are collected from the battery system. Then, key electrical and thermal features are extracted from the collected data through data preprocessing, feature monitoring, feature extraction and feature selection. Subsequently, fault diagnosis and fault prediction then occur based on the extracted features. Finally, battery faults are identified, isolated, and estimated. Additionally, fault prognosis can predict impending faults based on battery state estimations for preventative purposes.

A. Data Acquisition

Data acquisition refers to processing and storage of battery system data. In most researches, the battery data comes from experimental measurements and high-fidelity simulation models. Commonly collected data types include voltage, current, and temperature from respective sensors. For voltage measurement, three main measurement topologies exist, as shown in Fig. 8. Xia et al. [46] used the method shown in Fig. 8(a) to directly measure the battery voltage and calculated correlation coefficients to identify abnormal voltage changes.

In order to distinguish between battery and sensor faults, researchers [47], [48] employed the intercross measurement topology [Fig. 8(b)] to obtain battery voltage between positive and negative electrodes. For example, with a Cell 2 short circuit, the correlation coefficients $r(V_1, V_3)$ and $r(V_2, V_4)$ drop from 1 to a smaller value. When a connection fault occurs for Cell 1 and Cell 2, $r(V_1, V_2)$ and $r(V_3, V_4)$ drop at the same time. Xia and Mi [49] improved the intercross topology as shown in Fig. 8(c) which ensures that the voltage of each cell is associated with the measurements of two sensors.

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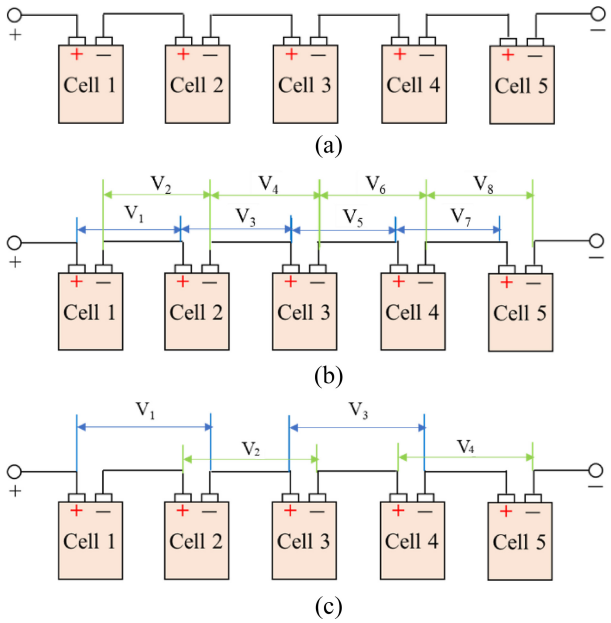


Fig. 8. Voltage measurement topology employed for battery fault diagnosis. (a) Conventional voltage measurement topology. (b) Cross voltage measurement topology 1 [48]. (c) Cross voltage measurement topology 2 [49].

For example, when Cell 2 is in external circuit condition, its terminal voltage drops to zero, revealed by decreases in V_1 and V_2. On the other hand, when there is a fault in V_2 voltage sensor, only V_2 will drop while all the other voltages remain the same.

B. Feature Extraction

After obtaining battery operation data, feature extraction is carried out on the raw data. Feature extraction includes the following three steps: 1) data preprocessing; 2) feature extraction; and 3) feature selection. Data preprocessing cleans and filters noise from the collected data for subsequent feature extraction and fault diagnosis. The commonly used noise removal methods include filtering and wavelet transforms. In [50], the voltage signal is preprocessed by wavelet transform before fault diagnosis. Since wavelet decomposition coefficients of the noise and the signal in different frequency bands have different intensity distributions, the wavelet coefficients corresponding to the noise in each frequency band were removed. Then, the processed coefficients were reconstructed by wavelet to obtain pure signals. In [51], the discrete cosine filtering method was adopted to optimize the truncation frequency according to the characteristics of white noise to achieve reasonable denoising.

Feature extraction is a key step in LIB fault diagnosis. Lithium-ion fault features refer to the characteristics or parameters that reflect the anomalies caused by faults. The fault can be detected by comparing extracted features of the target battery with thresholds or normal/average batteries. In general, LIB fault features come from three sources: 1) direct measurements, such as dV and dQ/dV; 2) battery model parameters, such as resistance (and its residue); and 3) signal features, such as root mean square, entropy, and outlier distance. Three common feature extraction methods are: 1) signal processing method; 2) model-based method; and 3) data-driven method.

1) Signal Processing Method: Signal processing method search for abnormalities in battery feature curves to capture the behavior of fault battery. Typical methods include correlation coefficient method, empirical mode decomposition (EMD) method, local outlier factor (LOF) method, entropy-based method, and principal component analysis (PCA).

The correlation coefficient method can be used to determine whether the trends of two curves match each other. This method reduces the impact of the inconsistency between open circuit voltage (OCV) and internal resistance on the accuracy of fault detection, and the detection process does not require complex analysis. In [52], batteries were arranged in ascending order of initial voltage, and every three batteries were divided into a group with two correlation coefficients calculated, thereafter voltage sensor faults, connection faults and short-circuit faults were detected according to the changes in correlation coefficients and voltage differences. Xie et al. [53] used recursive correlation coefficient to quantify the voltage difference between adjacent batteries, and then converted the sequence of correlation coefficients into false images for fault symptom analysis to detect the occurrence of faults, inferred fault types and evaluated fault grades. Li and Wang [54] proposed a new index called cross-correlation coefficient to identify the voltage drop that deviate from the voltage variation trend. This index can improve the accuracy of fault detection by amplifying the voltage difference.

For EMD, the signals collected by sensors are decomposed into natural mode functions of different frequencies. When a fault occurs, the phase frequency characteristics of the battery also change. By analyzing the corresponding frequency band, the fault cells can be identified. Li et al. [55] employed EMD to extract effective features from battery terminal voltage signals. However, EMD cannot effectively decompose signals containing substantial noise and inconsistencies, which can readily lead to mode mixing. Thus, Jiang et al. [56] processed voltage data by variational mode decomposition (VMD), extracting the static component highly related to aging inconsistency and dynamic component reflecting abnormal information to distinguish fault information hidden in individual inconsistencies. Jiang et al. [57] used wavelet transform to decompose the original voltage signal into low and high-frequency parts for separate fault characteristic extraction. However, VMD and wavelet transformation require subjective definition of parameters and are highly sensitive to parameter selection for different signals. Different VMDs and wavelet basis functions lead to great differences in feature extraction result. Thus, the two methods are difficult to be applied in practice.

Outlier detection involves finding anomalous behaviors compared to expected patterns through distance or density measurements. It does not need specific distribution of data, and can quantify the outlierness of each data point. Yuan et al. [58] proposed a local gravity anomaly detection method to evaluate voltage signals as fault indicators for early ISCs. In [50], LOF and Grubbs outlier filter were employed to identify cells with abnormal internal resistance. In [50], both discrete Fréchet distance and LOF were applied

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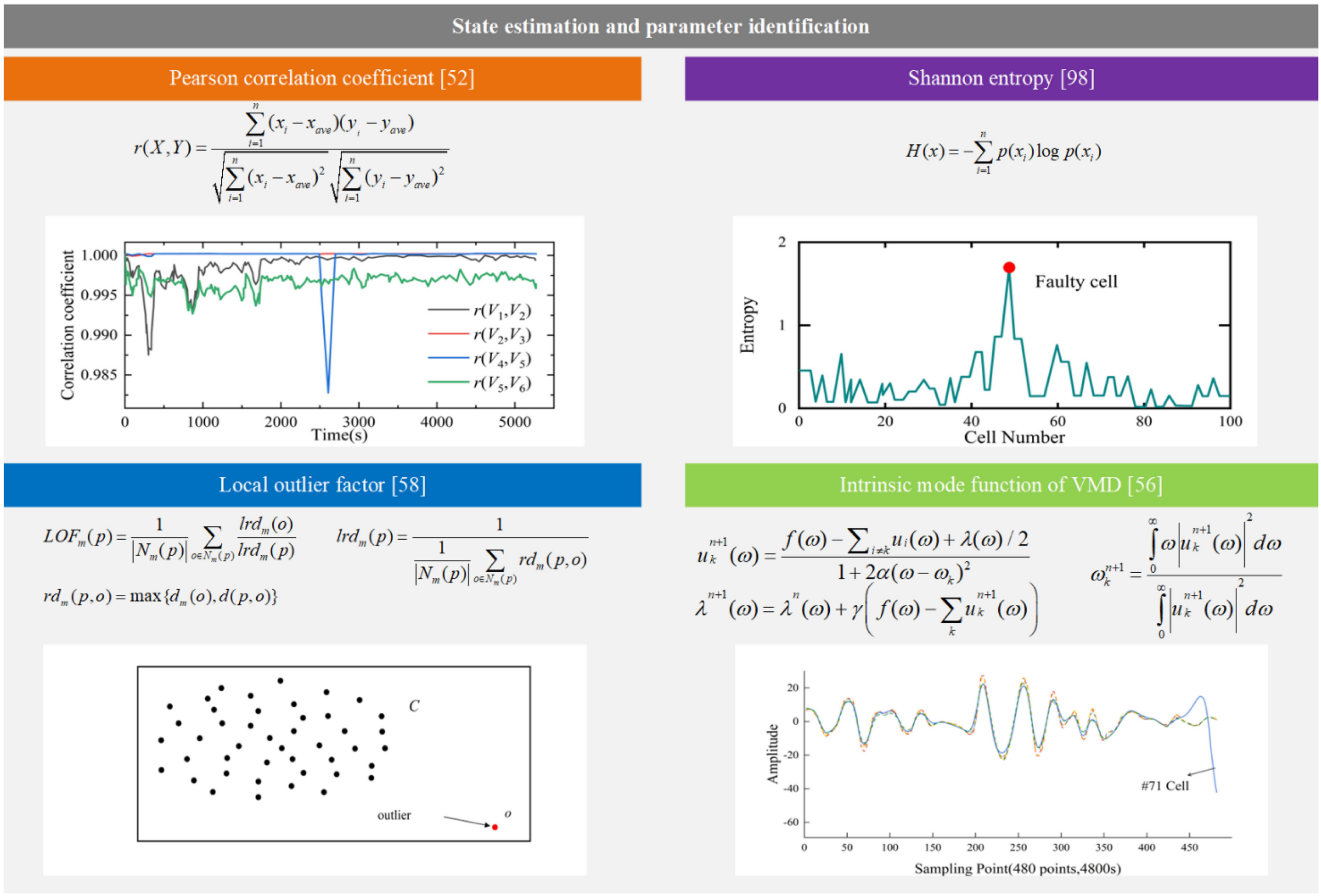


Fig. 9. Fault feature extraction with signal processing method.

to battery voltage and temperature to detect impending TR. However, these methods require calculating distances between data points, which computationally intensive.

Entropy-based method [60] can measure the degree of randomness or disorder of time series data to capture abnormal fluctuations. In [61], a multilevel concentration entropy battery fault diagnosis method was designed. The Shannon entropy of battery cell and the Shannon entropy of cluster were figured out for fault detection and the battery inconsistencies evaluation. Shang et al. [62] diagnosed and predicted different early battery failures by evaluating the modified sample entropy of the battery voltage sequence with a moving window.

Since many factors can induce battery faults, battery features tend to be mutually coupled. To reduce redundancy of these coupled features, methods like PCA [63], [64] are employed to reduce the dimension of features. In [65], PCA was used to decouple signals and isolate fault contributions by analyzing each element's contribution.

These methods are summarized in Fig. 9. In general, the signal processing method extracts useful fault features from the battery measurement data for fault detection without requiring accurate battery analysis model. Thus, signal processing method is suitable to diagnose multiple faults and to improve the overall diagnostic system performance. However, fault identification may not be possible since some battery failures have similar electrical and thermal responses. In addition, the signal processing method can only detect faults once the abnormal battery responses reach a certain level, making it difficult to detect small problems [4].

2) Model-Based Method: Parameters and states of battery models can also indicate faults. Compared to signal processing method, model-based method can accurately identify specific battery fault types. For example, a connection failure can cause a significant change in the contact resistance of the battery. ISC and thermal faults can be characterized by SoC reductions and ohm internal resistance increases, respectively.

According to the physical field, the models used for battery fault diagnosis include equivalent circuit model (ECM), thermal model, electrochemical model, and multiphysical field coupling model. When a battery fails, related model parameters and status of the battery change markedly. For example, after ISC occurs, the internal resistance of ISC becomes smaller [66]. Thus, some model parameters and states are important battery fault indicators. To evaluate battery states and parameters, filters [2], [2], [67], [68], observers [70], and parameter recognition algorithms, such as recursive least square (RLS) algorithm [71], particle swarm optimization algorithm [72], and genetic algorithm, are designed to apply on battery models.

ECM typically consists of an ideal voltage source representing OCV, RC loops representing dynamic characteristic of battery, and a resistor representing internal resistance. Current serves as the input and terminal voltage as the output. Generally, standard charging/discharging

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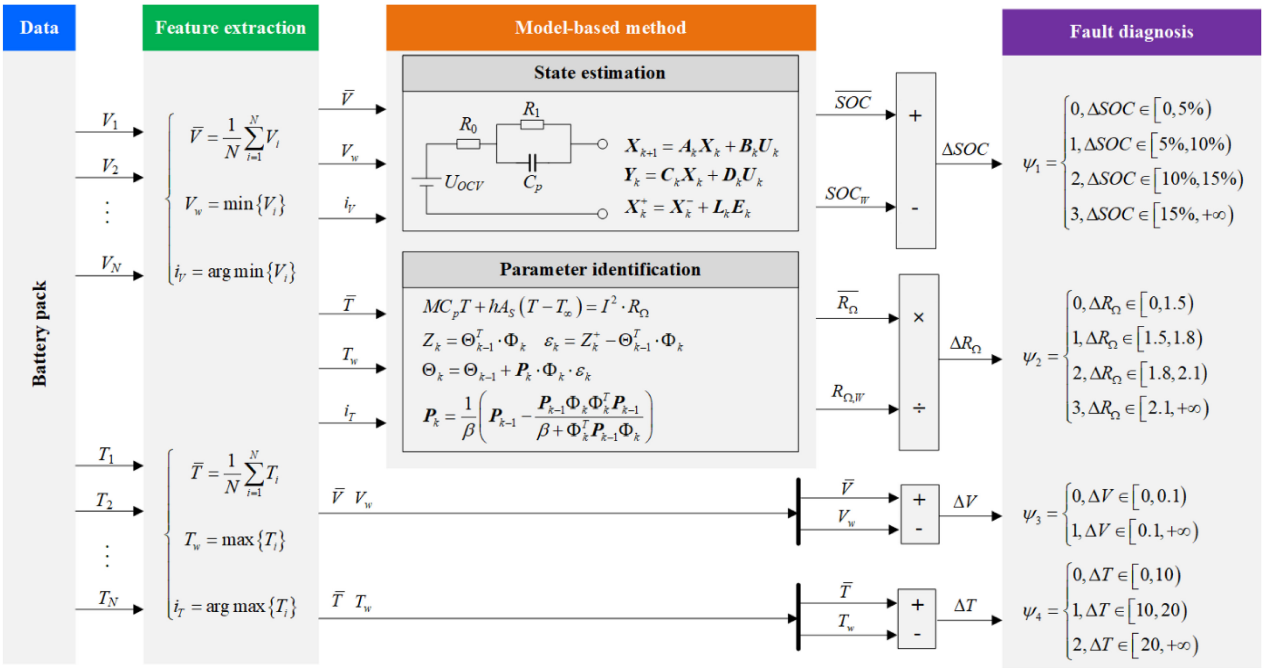


Fig. 10. Model-based battery fault diagnosis [79].

profiles like hybrid pulse power characterization (HPPC) and OCV tests are applied on offline parameter identification of ECM. Li et al. [73] provided an example of second order Thevenin ECM parameter identification using HPPC profile with detailed simulation process. To diagnosis certain fault like ISC, fault model like ISC model is established based on normal ECM and its parameters like ISC current and ISC resistance are identified as quantified indicator of ISC. In [74], an extended Kalman filter (EKF) was employed to estimate SoC, and to figure out ISC current. Based on the ISC current, a forgetting factor RLS method estimated ISC resistance as an important indicator of ISC. In [75], MSC failure was detected based on ECM. The MSC cell leakage current was obtained from the remaining charging capacity increases after each charge and converted into the MSC resistance for diagnosis. In [76], structural analysis method was employed to generate ECM residuals for detecting various sensor and cooling system faults. Besides single cell parameters and states, combined information from other cells in the battery pack, such as the differences between the faulty cell and the normal cells, can indicate faults. Yang et al. [77] compared voltage and temperature differences between faulty and normal batteries to thresholds for ISC detection. In [78], the SoC difference between worst and average batteries was obtained by interpolation of the OCV curve drawn out by EKF to calculate the soft short-circuit current and resistance. Feng et al. [79] provided a typical process of fault diagnosis with model-based method as shown in Fig. 10. Average and worst battery models were employed to estimate key parameters to diagnose multiple faults including ISCs. Mean-difference model (MDM) describes pack-level states for diagnosis [80]. In [70], MDM parameters were estimated by RLS, and then wave equations of features, such as voltage difference and internal resistance, were calculated to detect battery faults. These ECMs can be solved with MATLAB, Simulink and etc.

Thermal models calculate and predict cell temperatures, diagnosing faults from temperature behaviors [81]. In [82] and [83], convective cooling resistance fault, internal thermal resistance fault and TR were diagnosed using a two-state surface/core temperature observer model. Wei and Li [84] established a 2-D spatial model to simulate the temperature field and detected faults in both spatial and temporal domains. Li et al. [85] proposed a convolutional neural network (CNN)-long short-term memory (LSTM) neural network to predict temperature and identify abnormalities based on the thermal model. Thermal models solve in tools like Fluent, FloTherm, Icepak and etc.

Some studies couple multiphysical models to expand battery fault features. In [86], ISC and electrothermal coupling models were combined to calculate LOF and estimate the severity of ISC. In [87], an electrochemical-thermal coupling model was established, relating internal shorts to parameters for diagnosis. ECM, electrochemical model and thermal model are usually coupled as shown in Fig. 11 [87]. The boundary conditions are listed in the Appendix. Generally, electrochemical model and thermal model are solved with commercial software packages like COMSOL, LS DYNA, or ANSYS [88].

Besides these three models, some scholars also diagnose battery faults based on gas model, force model and sound model. When TR occurs in the battery, the sharp rise in temperature causes decomposition of internal substances, which produces gas leading to battery volume expansion. Cai et al. [89] coupled battery thermal model, expansion force model, and gas generation model to detect TR. Su et al. [90] proposed a battery fault diagnosis method based on the

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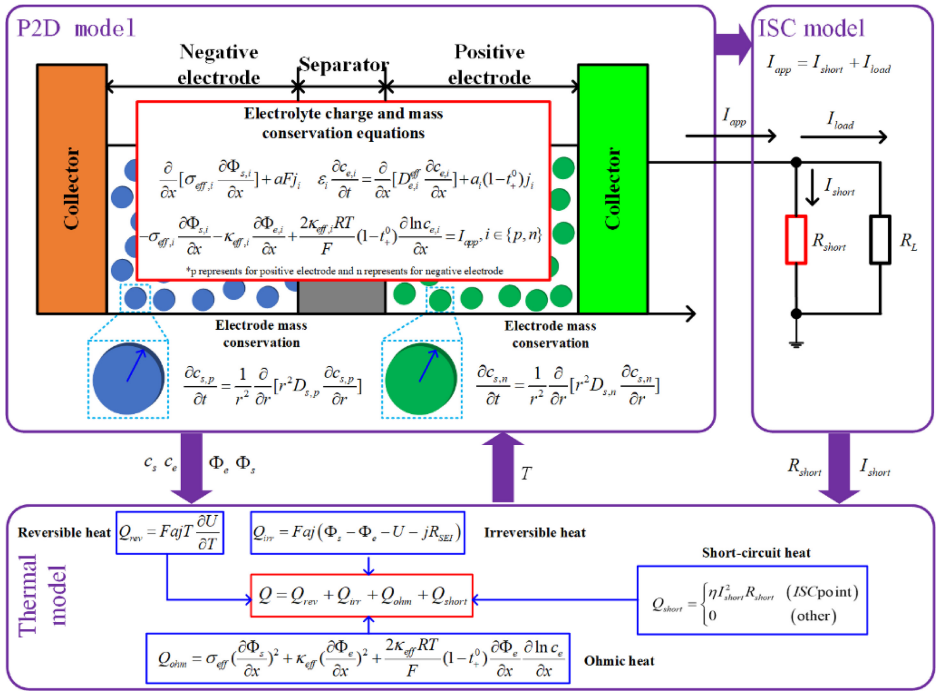


Fig. 11. Typical P2D-thermal-ISC coupled model [87].

battery exhaust sound signals in the energy storage system cabin. Through spectrum acquisition, ventilation noise, running noise, and typical interference noise in the battery cabin were analyzed and subtracted from the spectrum to eliminate the noise effects on TR diagnosis.

Overall, the model-based approach more readily quantifies and locates specific faults by exploiting relationships between the fault and the model states or parameters. However, fault diagnosis accuracy highly depends on model accuracy, state estimation and parameter identification accuracy.

3) Data-Driven Method: Data-driven algorithms offer an alternative to physics-based models for potentially more accurate fault feature extraction through online training and improvement, despite higher computational costs. Data driven methods like artificial neural network (ANN), LSTM, support vector machine (SVM), *K*-means are commonly used.

The term "Neural network" often referred to as ANN, is an important machine learning technology. Fig. 12 shows a typical neural network consisting of input layer, hidden layer and output layer. While input and output nodes are fixed, hidden nodes can vary. The connecting lines represent the connections between "neurons" and correspond to different weights that require training to be determined. LSTM is a type of neural network used for processing sequence data, which effectively solve the problems related to gradient vanishing and explosion that occur during long sequence training. In [91], LSTM was employed to predict current voltage based on inputs, such as cell voltage, SoC, and total voltage. Predicted current normal voltage, derived by coupling the voltage predictions from ECM and LSTM, was then used to identify voltage abnormalities. Similarly, in [85], a combination of CNN and LSTM was utilized with inputs, including vehicle state, driving behavior, and local weather data compressed through PCA, to predict temperature as Fig. 13 shows. Real world heat generation was figured out and compared with normal heat generation to forecast TR.

Besides various neural networks, classifiers like SVM and *K*-means can also be used for fault diagnosis. SVM was employed in [51] to optimize the parameters related to battery fault feature for fault identification. Xue et al. [92] described the parameter variation based on the key parameter variation of Gaussian distribution. *K*-means, on the other hand, was exploited to detect and locate the abnormal cells.

C. Fault Diagnosis

LIB fault diagnosis includes fault detection, fault isolation, fault estimation, and fault prognosis [93]. Fault detection refers to determine whether a fault has occurred. Fault isolation refers to the determination of fault type or location. Fault estimation is to assess the performance of fault diagnosis method. Fault prognosis is the early prediction of some battery faults with slow evolution [94].

1) Fault Detection: After extracting battery fault features, the fault is detected based on certain rules. Various methods are commonly employed for this purpose, including threshold-based method, residual-based method, statistic method, classification method, and joint method.

Threshold-based method is the most commonly used method. Certain battery parameters are closely related to specific faults, e.g., ISC resistance. Besides, deviations in feature values can indicate battery faults. Similarly, in [72], SoC difference served as a threshold for soft short circuits, validated through experimental testing. Wang et al. [95] employed Shannon entropy to monitor voltage fault, identifying the cell with the largest value of Shannon entropy regarded as faulted

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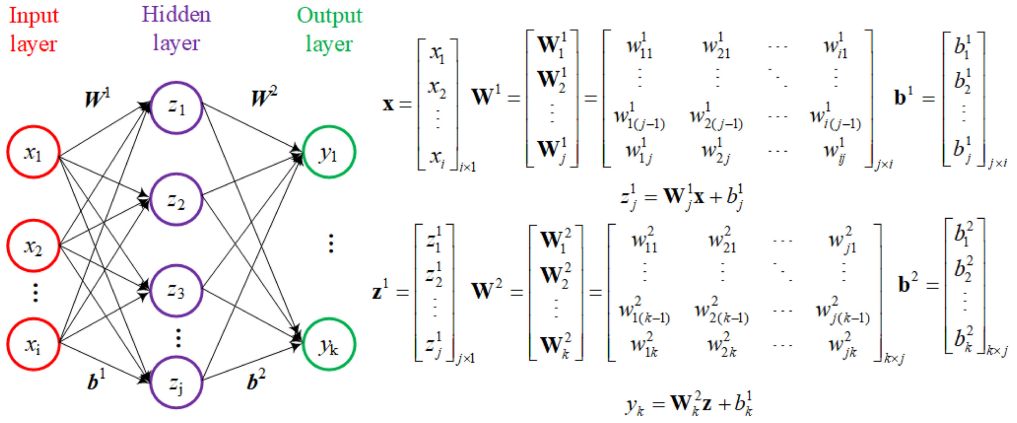


Fig. 12. Diagram of a typical neural network structure.

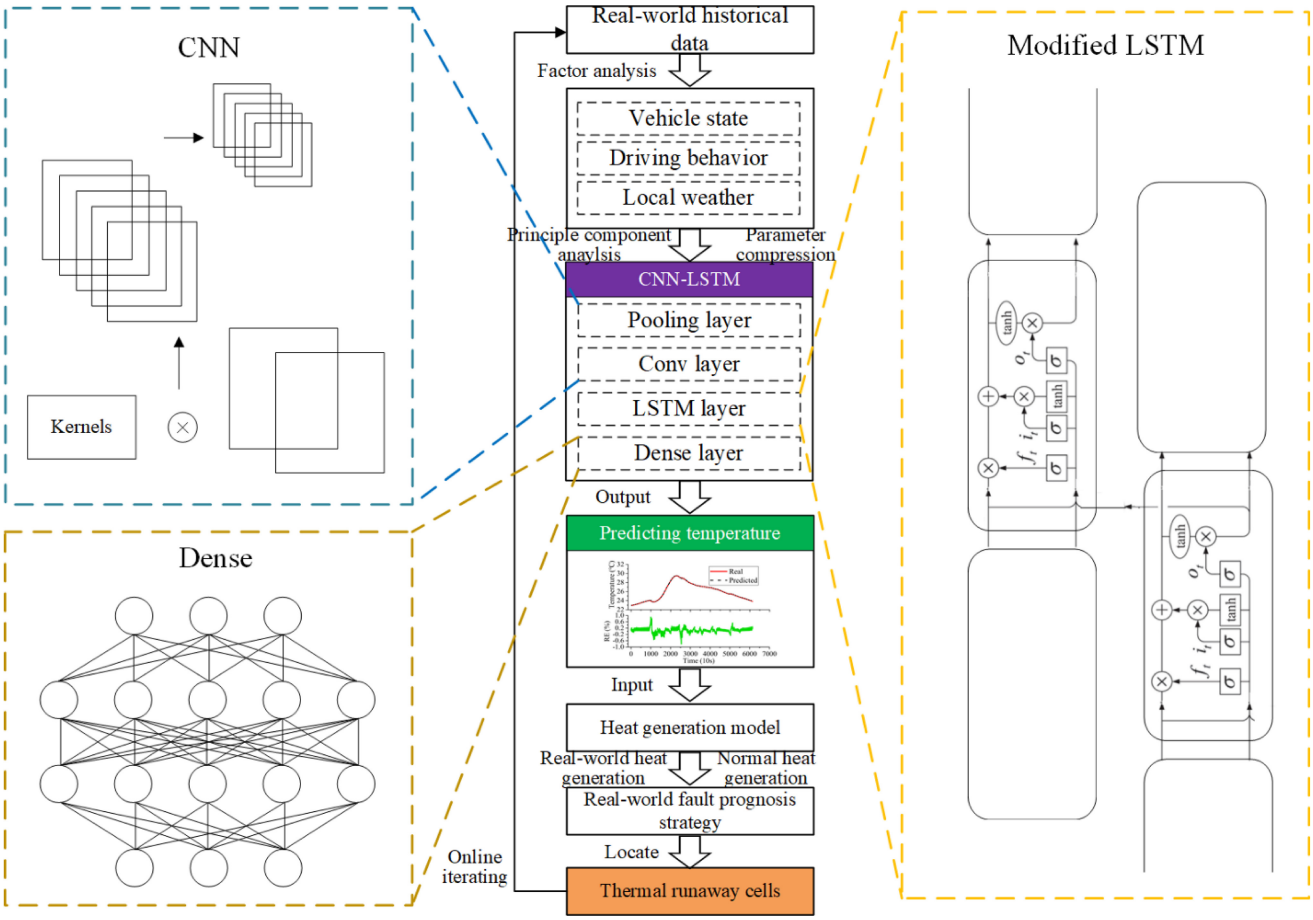


Fig. 13. CNN-LSTM-based fault diagnosis [85].

cell. While threshold-based detection method is convenient, setting appropriate thresholds is often relies on a large amount of experimental data and practical experience, which can be labor-intensive. In addition, model-based fault diagnosis methods require highly accurate battery model calculations and lack robustness against excitation or sensor noise, potentially leading to false diagnosis [96]. Nonmodel-based methods also struggle with threshold and parameter settings since they lack physical interpretations [82]. Based on battery model, residual-based criterion is another commonly approach for fault detection. Under normal conditions, the residual is zero, offering more flexibility compared to constant thresholds. Residuals are generated as the difference between measured value and output value of the model. In [97], residual of temperature and residual of voltage were evaluated based on ECM and thermal model to diagnosis battery fault. Both residuals transitioned up from zero to a nonzero value after the fault occurrence. In [98], cathode potential and anode potential were predicted to generate residuals using an electrochemical model. To mitigate the impact of noise, the variation of

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the residual curve under fault-free conditions is considered noise influence and removed to prevent incorrect actions. Dey et al. [83] considered residual response under no fault condition and regarded it as the bound of nonfault action. However, since these residual responses were measured by simulations or experiments with regular current profiles to avoid misdiagnosis, there is still possibility for wrong actions in harsh environmental and operational conditions.

In terms of statistical methods, the 3σ criterion is used to detect cells with significant voltage abnormalities that deviate significantly from the median cell voltage. In [99], the distribution of constant charging voltage ranking (CVR) variation was evaluated and the 3σ criterion was used to detect faulty cells with abnormal CVR variations. Zhao et al. [100] proposed a 3σ multilevel screening strategy to build the criteria of trouble-free cell terminal voltages. In order to eliminate the dimensional effects between feature indicators, features can be standardized by Z-score. Xue et al. [92] incorporated model-based method into Z-score method to detect battery abnormality. However, 3σ and Z-score methods assume that the data obey normal distribution while the actual data often do not strictly obey. Besides, their criteria for judging outliers rely on calculating the mean and standard deviation of the data batch, but outliers themselves will have a great impact on them, resulting in significant errors.

The severity of diagnosed battery faults can be identified by the deviation of features from these criteria.

2) Fault Isolation: Since the battery faults can be interrelated and may exhibit similar symptoms, such as overheating and voltage drops, it becomes necessary to identify the specific type of fault once it has been detected using methods like signal processing. Besides, some faults, like connection faults, can be challenging to locate, and the number of limit sensors also affect the ability to isolate a fault cell. Thus, effective methods for fault location are also necessary.

Many researchers have proposed multilevel diagnosis system to solve the problem. Sun et al. [101] proposed a multilevel diagnosis strategy to identify overvoltage faults, undervoltage faults, short circuits and open circuits. Besides, to isolate the battery faults when employing correlation coefficient diagnosis method, Li and Wang [54] converted correlation coefficient series into pseudo images with Gramian Angular Field and Markov Transition Field transformations to intuitively visualize the hidden cross-temporal autocorrelation features as image textures. Thereafter, the pseudo images were processed by CNN to infer fault type.

For some diagnosis methods that cannot pinpoint the fault's location, special measurement topology and sensor location optimization techniques are effective ways to identify fault battery. Xia and Mi [49] adopted measurement topology, as shown in Fig. 8(b), to identify the location and type of battery faults, like short circuits, sensor faults and connection faults. In [102], the temperature sensor locations were optimized to improve detectability and isolability of thermal faults.

3) Fault Estimation: The performance of a diagnosis system varies according to different methods employed. Thus, in order to design better fault diagnosis methods, it is critical to establish appropriate evaluation system for quantitatively comparing performance of different diagnosis methods [103].

According to different functions, the main performance indicators of the diagnosis system can be broadly divided into detection performance, diagnosis performance, and robustness [4]. Detection performance can be evaluated using sensitivity, detectable fault size, detection time, false alarm rate, missed detection rate, which are greatly affected by the threshold setting [104]. These indices demonstrate the promptness of battery fault handling. Diagnostic performance refers to the system's ability to isolate faults accurately and estimate fault characteristics [103]. The accuracy of fault estimation and isolation is closely tied to the chosen method and model accuracy. False alarm may lead to additional troubleshooting and security risks [4]. Robustness is the most difficult performance index to measure and realize. The robustness of diagnosis algorithm can be influenced by model uncertainty, disturbance, noise, and battery inconsistencies within a pack [53]. Kang et al. [105] proposed a framework for comparison of battery fault diagnostic methods. Sensitivity and robustness under different conditions of these diagnostic methods were quantitatively assessed and compared.

IV. Conclusion

This article provides a comprehensive survey of fault mechanisms, fault features, and fault diagnosis methods for LIBs. Due to the complexity of LIBs, no single battery fault diagnosis method is universally applicable to all types of battery faults. Fault diagnosis aims to detect the abnormalities caused by battery faults and determine the fault type. The fault diagnosis procedure typically includes four steps: 1) data acquisition; 2) feature extraction; 3) fault diagnosis; and 4) fault prognosis.

In terms of data acquisition, the signals used for fault diagnosis typically include temperature, current, and voltage obtained from sensors. These external behaviors show abnormalities when battery faults occur and can be used to estimate battery parameters. Other signals obtained from gas sensors and force sensors are rarely considered.

Feature extraction is the most critical step in the diagnosis process. Battery features reflecting fault information can be classified as features with specific physical meaning obtained by direct measurements and model-based methods, as well as features without specific physical meaning obtained by signal processing method, machine learning method, and stochastic method. Signal processing method is widely applied due to its low-computational burden and high generality, making it suitable for multifault diagnosis. However, it can be challenging to differentiate between battery fault types with features extracted through signal processing method, since many faults exhibit similar symptoms. Model-based method can extract parameters reflecting battery internal characteristics with physical meaning that are closely related to certain faults, but the accuracy of established model influences the feature extraction accuracy. Data-driven method can establish relationship between battery behavior and certain

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faults avoiding specific physical model, while it suffers low robustness, high-computation costs, and dependency on high-quality training data.

After extracting fault features, battery fault diagnosis involves three main steps: 1) fault detection; 2) fault isolation; and 3) fault estimation. Fault detection determines whether a battery fault has occurred, typically using threshold-based or residual-based criteria. However, selecting appropriate thresholds and avoiding wrong actions based on residuals remain challenges, as these factors influence the balance between sensitivity and robustness, which in turn affects diagnosis accuracy. Fault isolation aims to identify the specific type and location of a battery fault. Multilevel fault diagnosis schemes are usually designed to figure out battery type by setting various judgment criteria. Special measurement topologies and sensor location optimization techniques are useful to locate faulty battery. Fault estimation evaluates the performance of the fault diagnosis method, like detection performance, diagnosis performance, and robustness, which is an important step to design effective diagnosis system.

Although many researchers have done a lot of work in battery fault diagnosis, most developed methods face practical challenges.

1) Methods are often developed based on experiments conducted under fixed working conditions and mild environments, limiting their applicability in industrial settings.
2) Batteries are usually assembled in packs and in clusters to provide higher power, increasing system complexity, which makes fault diagnosis difficult. Existing approaches are often developed with limited battery samples and lack systematic diagnosis strategies.
3) Current methods are usually designed for single faults, yet various fault types exist, making it impossible to provide each fault type with a unique diagnosis strategy.
4) Current methods mainly rely on voltage, current, and temperature signals, which is not sufficient for battery fault diagnosis, as it may not exhibit abnormal behavior before some faults occur.

It is clear that more and more researchers begin to diagnosis different faults at the same time and identify different faults with similar behaviors by simple rules. Battery fault mechanism is also studied more to figure out more internal features as fault indicators.

Thus, further research in LIB fault diagnosis is needed, focusing on the following.

1) Developing diagnosis methods suitable for practical applications.
2) Conducting multiscale mechanism studies at cell, pack, and system levels.
3) Investigating multifault diagnostics.
4) Exploring approaches that consider multiple physical fields.

The complexity and importance of LIB fault diagnosis underscore the need for ongoing research and innovation in this field.

APPENDIX

Governing equations and boundary conditions of the P2D model.

Governing Equations	Boundary Conditions
Positive and negative electrode	
	$\frac{\partial c}{\partial x}\big\|_{x=0}=0$
	$-D_{eff,p}\frac{\partial c}{\partial x}\Big\|_{x=l_p^-}=-D_{eff,s}\frac{\partial c}{\partial x}\Big\|_{x=l_p^+}$
	$\frac{\partial c}{\partial x}\Big\|_{x=l_p+l_s+l_n}=0$
$\varepsilon_i\frac{\partial c_{e,i}}{\partial t}=\frac{\partial}{\partial x}[D_{e,i}^{eff}\frac{\partial c_{e,i}}{\partial x}]+a_i(1-t_+^0)j_i, i\in\{p,n\}$	$-D_{eff,s}\frac{\partial c}{\partial x}\Big\|_{x=l_p+l_s^-}=-D_{eff,n}\frac{\partial c}{\partial x}\Big\|_{x=l_p+l_s^+}$
$\frac{\partial c_{s,i}}{\partial t}=\frac{1}{r^2}\frac{\partial}{\partial r}[r^2D_{s,i}\frac{\partial c_{s,i}}{\partial r}], i\in\{p,n\}$	$\frac{\partial c_{s,i}}{\partial r}\Big\|_{r=0}=0$
$-\sigma_{eff,i}\frac{\partial\Phi_{s,i}}{\partial x}-\kappa_{eff,i}\frac{\partial\Phi_{e,i}}{\partial x}+\frac{2\kappa_{eff,i}RT}{F}(1-t_+^0)\frac{\partial\ln c_{e,i}}{\partial x}$ $=I_{app}, i\in\{p,n\}$	$D_{s,i}\frac{\partial c_{s,i}}{\partial r}\Big\|_{r=R_i}=j_i, i\in\{p,n\}$
	$\frac{\partial\Phi_{e,p}}{\partial x}\Big\|_{x=0}=0\ \Phi_{e,n}\Big\|_{x=l_p+l_s+l_n}=0$
$\frac{\partial}{\partial x}[\sigma_{eff,i}\frac{\partial\Phi_{s,i}}{\partial x}]+aFj_i, i\in\{p,n\}$	$-\kappa_{eff,p}\frac{\partial\Phi_{e,p}}{\partial x}\Big\|_{x=l_p^-}=-\kappa_{eff,s}\frac{\partial\Phi_{e,p}}{\partial x}\Big\|_{x=l_p^+}$
$j_i=2k_ic_{e,i}^{0.5}c_{s,i}\Big\|_{r=R_i}^{0.5}(c_{\max,i}^s-c_{s,i}\Big\|_{r=R_i})^{0.5}$	$-\kappa_{eff,s}\frac{\partial\Phi_{e,s}}{\partial x}\Big\|_{x=l_p+l_s^-}=-\kappa_{eff,n}\frac{\partial\Phi_{e,n}}{\partial x}\Big\|_{x=l_p+l_s^+}$
$\sinh[\frac{F}{2RT}(\Phi_{s,i}-\Phi_{e,i}-U_{OCV,i})], i\in\{p,n\}$	$\frac{\partial\Phi_{s,p}}{\partial x}\Big\|_{x=0}=\frac{I_{app}}{\sigma_{eff,p}}$
	$\frac{\partial\Phi_{s,n}}{\partial x}\Big\|_{x=l_p+l_s+l_n}=-\frac{I_{app}}{\sigma_{eff,n}}$
	$\frac{\partial\Phi_{s,p}}{\partial x}\Big\|_{x=l_p^-}=0$
	$\frac{\partial\Phi_{s,n}}{\partial x}\Big\|_{x=l_p+l_s^-}=0$
Separator	
	$c_{e,p}\Big\|_{x=l_p^-}=c_{e,s}\Big\|_{x=l_p^+}$
$\varepsilon_s\frac{\partial c_{e,s}}{\partial t}=\frac{\partial}{\partial x}[D_{e,s}^{eff}\frac{\partial c_{e,s}}{\partial x}]$	$c_{e,s}\Big\|_{x=l_p+l_s^-}=c_{e,n}\Big\|_{x=l_p+l_s^+}$
$-\kappa_{eff,s}\frac{\partial\Phi_{e,s}}{\partial x}+\frac{2\kappa_{eff,s}RT}{F}(1-t_+^0)\frac{\partial\ln c_{e,s}}{\partial x}=I_{app}$	$\Phi_{e,s}\Big\|_{x=l_p^-}=\Phi_{e,p}\Big\|_{x=l_p^+}$
	$\Phi_{e,s}\Big\|_{x=l_p+l_s^-}=\Phi_{e,n}\Big\|_{x=l_p+l_s^+}$

The technical specifications of battery cell used in the reference of model-based method.

Ref.	Type	Voltage range	Nominal voltage	Nominal capacity
[67]	NCM		3.65 V	2.85 Ah
	LFP		3.2 V	20 Ah
[68]	LTO		2.4 V	10 Ah
	NCM		3.6 V	20 Ah
[69]	NCM			
[72]	LFP		3.3 V	2.3 Ah
[73]	NCM			20 A
[74]	NCM		3.6 V	2.0 Ah
[75]	NCM		3.6 V	5 Ah
[76]	NCM			2.2 Ah
[77]	NCM			120 Ah
[79]	NCM			5 Ah
[80]	NCM	2.75 V - 4.2 V		25 Ah
[82]	NCM	3 V - 4.18 V		190 Ah
	NCM	2.5 V - 4.2 V		158 Ah
[82]	LFP	2.5 V - 3.65 V		250 Ah
[86]			3.2 V	20 Ah
[87]				28 Ah
[88]	LFP		3.2 V	40 Ah
[89]	NCM			27 Ah
[70], [71], [78], [81], [84], [85] do not mention				

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